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ACADEMIC VAULT

UNIZIK FRESHER EDITION | 2026

MTH 101: Elementary Mathematics I - Algebra, Trigonometry and Sequences

Close the secondary-school gaps that make first-year mathematics feel impossible

FACULTY PRIMER

Faculty Primer

Faculty of Physical Sciences
Mathematics | 100L

Written for students, by students.

Practical notes, worked examples, practice questions and a revision plan.

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Independent study support | Not an official UNIZIK publication

START HERE

Why this pack exists

MTH 101 assumes you already own certain algebraic habits. Many freshers do not, and the course moves fast. This pack rebuilds those habits - factorisation, indices, quadratic theory, trigonometric identities and sequences - with short explanations and worked examples that model good layout.

This is a student-written study companion, not a reproduced lecture handout or a leaked examination paper. Use it to organise the topics you are taught, practise the kind of thinking your course rewards, and walk into class with better questions.

What you will be able to do

- Factorise and simplify algebraic expressions including surds and partial fractions.
- Solve quadratic equations and use the sum and product of roots.
- Apply trigonometric ratios, identities and the sine and cosine rules.
- Work confidently with arithmetic and geometric progressions, including sums to n terms.

How to use this pack

1. Skim the whole pack once in 20 minutes so you know what is inside.
2. Work through one section at a time with your lecture note and recommended text beside you.
3. Attempt the practice questions before reading the answers. Mark what you missed.
4. Use the final checklist to plan your revision week, one tick per study session.

> **Free to download, always**
> This pack is free. There is no fee, no subscription and no payment step before download, and the full PDF is not watermarked or locked. Share the link with a coursemate instead of forwarding the file.

> **Confirm your official course outline**
> Course codes, credit units and outlines can differ by department and session. Treat this pack as supplementary support and confirm the current outline with your lecturer, course rep or departmental handbook before an assessment.

Algebraic fluency

Speed and accuracy in algebra decide your grade more than clever tricks do.

- Know the difference of two squares, perfect squares and the factor theorem.
- Rationalise surds by multiplying by the conjugate.
- Solve simultaneous equations by substitution when one equation is linear.
- For quadratics, use factorisation first, completing the square second and the formula last.
- For ax^2 + bx + c = 0, the sum of roots is -b/a and the product is c/a.
- Check answers by substitution - it takes seconds and catches sign errors.

> **Show your line of reasoning**
> Marks are awarded for correct intermediate steps. A bare answer that is wrong earns nothing; a clear method with a slip often earns most of the marks.

Trigonometry that stays consistent

Trigonometry rewards a diagram and punishes memorised ratios applied blindly.

- Use the unit circle to determine the sign of each ratio in each quadrant.
- Sine rule: a/sin A = b/sin B = c/sin C.
- Cosine rule: a^2 = b^2 + c^2 - 2bc cos A.
- Area of a triangle: half ab sin C.
- Core identities: sin^2 + cos^2 = 1; tan = sin/cos; double-angle formulae.
- Convert between degrees and radians before using calculus formulae.

Sequences and series

Identify the type of sequence before touching a formula.

- Arithmetic progression: common difference d; nth term a + (n-1)d; sum n/2(2a + (n-1)d).
- Geometric progression: common ratio r; nth term ar^(n-1); sum a(1 - r^n)/(1 - r) for r not equal to 1.
- Sum to infinity exists only when the absolute value of r is less than 1, and equals a/(1 - r).
- For word problems, write the first three terms explicitly to confirm the pattern.
- Binomial expansion: use Pascal's triangle for small powers, the general term for larger ones.

PRACTICE, NOT PREDICTION

Check your understanding

These questions are written for this pack to test the ideas above. They are not past examination questions and do not predict what your lecturer will set. Answer first, then compare.

1. Solve x^2 - 5x + 6 = 0 and state the sum and product of the roots.

Answer: Factorising gives (x - 2)(x - 3) = 0, so x = 2 or 3. Sum of roots is 5 and product is 6.

2. Find the 10th term of the AP 3, 7, 11, ...

Answer: a = 3, d = 4, so the 10th term is 3 + 9(4) = 39.

3. Find the sum to infinity of 8, 4, 2, ...

Answer: a = 8, r = 0.5, so sum = 8/(1 - 0.5) = 16.

4. In a triangle, b = 7, c = 9 and angle A = 60 degrees. Find a.

Answer: a^2 = 49 + 81 - 2(7)(9)(0.5) = 130 - 63 = 67, so a is about 8.19.

5. Simplify 1/(sqrt(3) - 1) by rationalising.

Answer: Multiply numerator and denominator by sqrt(3) + 1 to get (sqrt(3) + 1)/2.

Your 7-step action plan

Tick each step only when you have done it. A tick you cannot defend is worse than an empty box.

- [ ] Write out the factorisation identities and solve ten expressions using them.
- [ ] Derive the quadratic formula by completing the square.
- [ ] Draw the unit circle and mark the signs of sine, cosine and tangent in each quadrant.
- [ ] Solve five triangle problems using the sine and cosine rules.
- [ ] Work through three AP and three GP problems including sum to infinity.
- [ ] Do a timed 30-question arithmetic-and-algebra drill.
- [ ] Review every wrong answer and name the specific rule you misapplied.

Keep going

One pack will not carry a semester. Pair this with your lecture notes, tutorials, and the past questions your department actually releases. If a topic still feels unclear, take a specific question to your lecturer or course rep - "I do not understand the difference between X and Y" gets a better answer than "I do not understand this course".

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YOUR WORKSPACE

Notes, questions and things to ask

Use this page in class or during revision. Write questions you could not answer instead of copying slides.

Three ideas I can now explain without notes:

A question I still cannot answer:

For my lecturer or course rep:

Next revision action and date:

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